

Name: _____ Class: _____

WHITEBRIDGE HIGH SCHOOL



2007

Trial HSC Examination

MATHEMATICS EXTENSION 2

Time Allowed: Two and a half hours

(Reading time: 5 minutes)

Directions to Candidates

- All questions to be completed on writing paper provided
- Commence each question on a new page.
- Marks may be deducted for careless or badly arranged work.
- Calculators may be used.

Question 1 (15 marks) Commence each question on a SEPARATE page

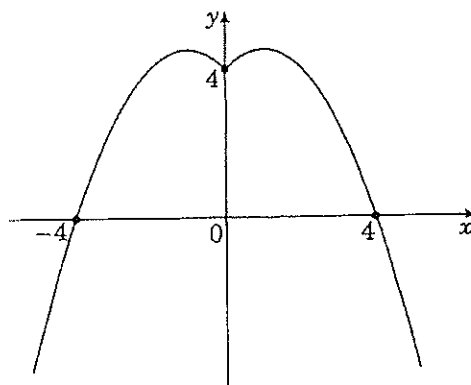
Marks

- a. Find $\int \frac{\sin x}{\cos^3 x} dx$. **2**
- b. By completing the square, evaluate $\int \frac{1}{x^2 + 2x + 5} dx$. **2**
- c. (i) Given that $\frac{4x - 6}{(x + 1)(2x^2 + 3)}$ can be written as **3**
$$\frac{4x - 6}{(x + 1)(2x^2 + 3)} = \frac{a}{x + 1} + \frac{bx + c}{2x^2 + 3},$$
 where a , b and c are real numbers,
find a , b and c .
- (ii) Hence find $\int \frac{4x - 6}{(x + 1)(2x^2 + 3)} dx$. **2**
- d. Evaluate $\int_0^1 x \tan^{-1} x \, dx$. **3**
- e. Use the substitution $t = \tan \frac{\theta}{2}$ to show that $\int_0^{\frac{\pi}{3}} \frac{d\theta}{13 + 5 \sin \theta + 12 \cos \theta} = \frac{2}{5 + 25\sqrt{3}}$. **3**

Question 2 (15 marks) Commence each question on a SEPARATE page

Marks

- a. The even function $y = f(x)$ is shown below.



On a separate number plane sketch each of the following, showing all important features.

(i) $y = f(x - 4)$ 2

(ii) $y = |f(x)|$ 2

(iii) $y^2 = f(x)$ 2

- b. The function $y = f(x)$ is defined by $f(x) = \frac{\ln x}{x}$, for $x > 0$.

(i) Determine any stationary points, points of inflexion and equations of possible asymptotes for $y = f(x)$. 4

(ii) Draw a sketch of $y = f(x) = \frac{\ln x}{x}$, showing all relevant details determined from (i). 1

(iii) Draw separate sketches of the graphs of

(α) $y = \left| \frac{\ln x}{x} \right|$ 2

(β) $y = \frac{x}{\ln x}$ 2

Question 3 (15 marks) Commence each question on a SEPARATE page

Marks

a. Let $z = 5 + 3i$ and $w = -3 + 2i$.Express the following in the form $a + bi$, where a and b are real numbers.

(i) $\overline{zw}$

1

(ii) $\frac{2}{iw}$

1b. (i) Express $\sqrt{3} + i$ in mod-arg form.**2**(ii) Find the exact value of $(\sqrt{3} + i)^{12}$ in the form $a + bi$ where a and b are real numbers.**2**

c. On separate diagrams, sketch the region where the inequalities hold:

(i) $|z - 3 + i| \leq 5$ and $|z + 2| \leq |z - 2|$.

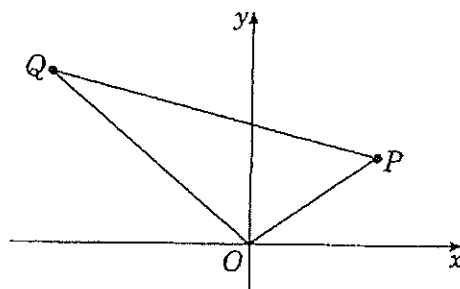
2

(ii) $2 < |z| < 3$ and $\frac{\pi}{6} < \arg z < \frac{\pi}{2}$.

2d. (i) If $w = \frac{1 + \sqrt{3}i}{2}$, show that $w^3 = -1$.**1**(ii) Hence calculate w^{16} .**1****Question 3 continues over page**

Question 3 continued

e.



The diagram shows a complex plane with origin O.

The points P and Q represent arbitrary non-zero complex numbers z_1 and z_2 respectively.

Thus the length of PQ is $|z_1 - z_2|$.

- (i) Use the diagram to show $|z_1 - z_2| \leq |z_1| + |z_2|$. **1**
- (ii) Construct the point R representing $z_1 + z_2$. **1**
What type of quadrilateral is OPRQ?
- (iii) If $|z_1 - z_2| = |z_1 + z_2|$, what can be said about the complex number $\frac{z_2}{z_1}$? **1**

Question 4 (15 marks) Commence each question on a SEPARATE page

Marks

- a. The roots of $x^3 + 5x^2 + 11 = 0$ are α , β and γ .
- (i) Find the polynomial equation whose roots are α^2 , β^2 and γ^2 . **2**
- (ii) Find the value of $\alpha^2 + \beta^2 + \gamma^2$. **1**
- b. (i) Suppose the polynomial $P(x)$ has a double root at $x = a$.
Prove that $P'(x)$ also has a root at $x = a$. **1**
- (ii) The polynomial $P(x) = x^4 + ax^3 + bx + 21$ has a double root at $x = 1$.
Find the values of a and b . **2**
- (iii) Factorise the polynomial $P(x)$ of part (i) over the real numbers. **2**
- c. Factorise $P(x) = x^4 - 5x^3 + 7x^2 + 3x - 10$ over $\mathbb{C}$ if $2 - i$ is a zero. **3**
- d. (i) Derive the five roots of the equation $z^5 - 1 = 0$. **2**
- (ii) Hence find the exact value of $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5}$. **2**

Question 5 (15 marks) Commence each question on a SEPARATE page

Marks

a. The ellipse E has the equation $4x^2 + 9y^2 = 36$.

(i) Write down:

(α) its eccentricity 1

(β) the coordinates of its foci S and S' . 1

(γ) the equation of each directrix 1

(δ) the length of the major axis. 1

(ii) Sketch the ellipse E . 1

Show the x and y intercepts as well as the features found in parts (β) and (γ) of part (i) above.

b. H is a rectangular hyperbola whose equation is given by $xy = \frac{1}{2}a^2$.

(i) Prove that, for all values of t , the point $P(\frac{at}{2}, \frac{a}{t})$ lies on H and that the equation of the tangent at P has the equation $2x + t^2y = 2at$. 2

(ii) S is the point (a, a) and the perpendicular from S to the tangent described in (i) above meets the tangent at T . 2
Prove that $t^2x - 2y = at^2 - 2a$ is the equation of the line ST .

(ii) Show that, as P moves on the hyperbola, the locus of T is a circle. 2

Question 5 continues over page

Question 5 continued

c. The point $P(a \sec \theta, b \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

(i) Show that the equation of the tangent to the hyperbola at P is given

3

by $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$.

(ii) The tangent to the hyperbola at P meets the asymptotes at A and B.
Prove that P is the midpoint of the interval AB.

3

Question 6 (15 marks) Commence each question on a SEPARATE page

Marks

- a. (i) Find the five fifth roots of $\sqrt{3} + i$. **2**
- (ii) Show the roots on an Argand diagram. **2**
- (iii) Find the area of the pentagon formed by the roots. **1**
- b. (i) Given that $C_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, prove that $C_n = \frac{n-1}{n} C_{n-1}$, where $n = 2, 3, 4, 5, \dots$ **3**
- (ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^4 x \, dx$. **1**

Question 6 continues over page

Question 6 continued

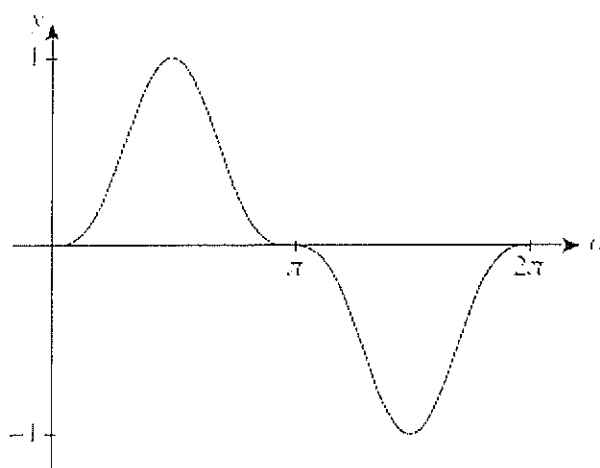
c. Consider the complex number $z = \cos \alpha + i \sin \alpha$

(i) Prove that $z^n - \frac{1}{z^n} = 2i \sin n\alpha$. **2**

(ii) Expand $(z - \frac{1}{z})^3$, and use the result from part (i) to show **2**

$$\text{that } \sin^3 \alpha = \frac{3}{4} \sin \alpha - \frac{1}{4} \sin 3\alpha.$$

(iii) Below is a sketch of $y = \sin^3 \alpha$ for $0 \leq \alpha \leq 2\pi$. **2**



Find the area of the region between the curve and the α -axis, for $0 \leq \alpha \leq 2\pi$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note: $\ln x = \log_e x, \quad x > 0$

Question 1:

a. $\int \frac{\sin x}{\cos^3 x} dx$

Let $u = \cos x$

$\frac{du}{dx} = -\sin x$

$dx = \frac{du}{-\sin x}$

$= \int \frac{\sin x}{u^3} \cdot \frac{du}{-\sin x}$

$= - \int u^{-3} du$

$= - \frac{u^{-2}}{-2}$

$= \frac{1}{2u^2}$

$= \frac{1}{2\cos^2 x} + C$

(2)

b. $\int \frac{1}{x^2+2x+5} dx$

$= \int \frac{1}{x^2+2x+1+4} dx$

$= \int \frac{1}{(x+1)^2+2^2} dx$

$= \frac{1}{2} \tan^{-1}\left(\frac{x+1}{2}\right) + C$

(2)

c. i. $\frac{4x-6}{(x+1)(2x^2+3)} = \frac{a}{(x+1)} + \frac{bx+c}{(2x^2+3)}$

$= \frac{a(2x^2+3) + (bx+c)(x+1)}{(x+1)(2x^2+3)}$

$\therefore 4x-6 = a(2x^2+3) + (bx+c)(x+1)$

Subs $x = -1$

$\therefore -10 = 5a \therefore a = -2$

Subs $x = 0$

$\therefore -6 = 3a + c(1)$

$\therefore -6 = -6 + c \therefore c = 0$

(3)

Subs $x = 1$

$-2 = 5a + (b+c) \cdot 2$

$\therefore -2 = -10 + 2b$

$2b = 8 \therefore b = 4$

$\therefore a = -2$
 $b = 4$
 $c = 0$

ii. $\int \frac{-2}{x+1} dx + \frac{4x}{2x^2+3} dx$

$= -2 \ln(x+1) + \ln(2x^2+3) + C$

(2)

d. $\int_0^1 x \tan^{-1} x dx$

$u = \tan^{-1} x$

$u' = \frac{1}{1+x^2}$

$v' = x$

$v = \frac{x^2}{2}$

$= \left[\frac{x^2}{2} \tan^{-1} x \right]_0^1 - \int_0^1 \frac{1}{1+x^2} \cdot \frac{x^2}{2} dx$

$= \left(\frac{1}{2} \cdot \frac{\pi}{4} - 0 \right) - \frac{1}{2} \int_0^1 \frac{x^2}{1+x^2} dx$

$= \frac{\pi}{8} - \frac{1}{2} \int_0^1 \frac{1+x^2-1}{1+x^2} dx$

$= \frac{\pi}{8} - \frac{1}{2} \int_0^1 \left(1 - \frac{1}{1+x^2} \right) dx$

$= \frac{\pi}{8} - \frac{1}{2} \left[x - \tan^{-1} x \right]_0^1$

$= \frac{\pi}{8} - \frac{1}{2} \left[1 - \tan^{-1} 1 - 0 \right]$

$= \frac{\pi}{8} - \frac{1}{2} \left[1 - \frac{\pi}{4} \right]$

$= \frac{\pi}{8} - \frac{1}{2} + \frac{\pi}{8} = \frac{\pi}{4} - \frac{1}{2}$

(2)

(1)

e. $\int_0^{\pi/3} \frac{d\theta}{13+5\sin\theta+12\cos\theta}$

$t = \tan \frac{\theta}{2}$

Now, $13+5\sin\theta+12\cos\theta$

$= 13 + 5 \left[\frac{2t}{1+t^2} \right] + 12 \left[\frac{1-t^2}{1+t^2} \right]$

$= 13 + \frac{10t}{1+t^2} + \frac{12-12t^2}{1+t^2}$

$= \frac{13+13t^2+10t+12-12t^2}{1+t^2}$

$= \frac{t^2+10t+25}{1+t^2} \quad d\theta = \frac{2 dt}{1+t^2}$

$\therefore \int_0^{\frac{1}{\sqrt{3}}} \frac{1+t^2}{t^2+10t+25} \cdot \frac{2 dt}{1+t^2}$

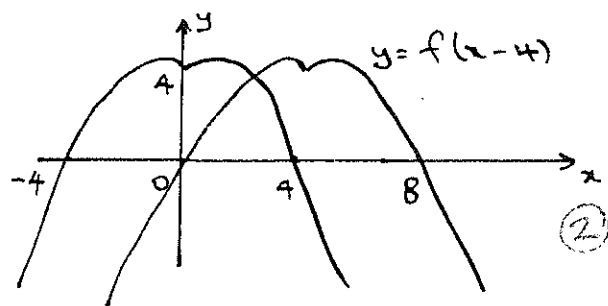
$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{t^2+10t+25} dt$

$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{(t+5)^2} dt$

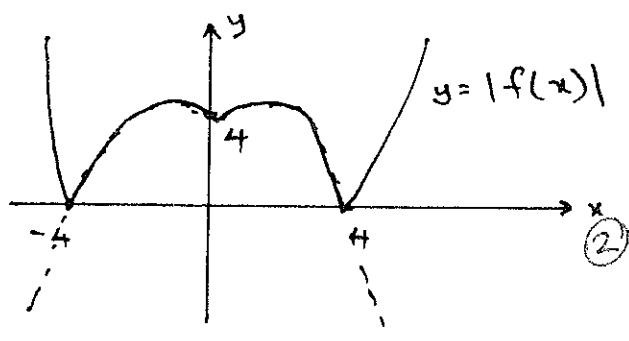
$$\begin{aligned}
 &= 2 \int_0^{\sqrt[3]{3}} (t+5)^{-2} dt \\
 &= 2 \left[\frac{(t+5)^{-1}}{-1} \right]_0^{\sqrt[3]{3}} \\
 &= \left[\frac{-2}{(t+5)} \right]_0^{\sqrt[3]{3}} \\
 &= \left[\frac{-2}{\sqrt[3]{3}+5} - \frac{-2}{5} \right] \times \frac{\sqrt[3]{3}}{\sqrt[3]{3}} \\
 &= \frac{-2\sqrt[3]{3}}{1+5\sqrt[3]{3}} + \frac{2\sqrt[3]{3}}{5\sqrt[3]{3}} \\
 &= \frac{-20+2\sqrt[3]{3}+20}{5\sqrt[3]{3}(5\sqrt[3]{3}+1)} = \frac{2}{5(5\sqrt[3]{3}+1)} \\
 &= \frac{2}{25\sqrt[3]{3}+5} \quad (3)
 \end{aligned}$$

Question 2

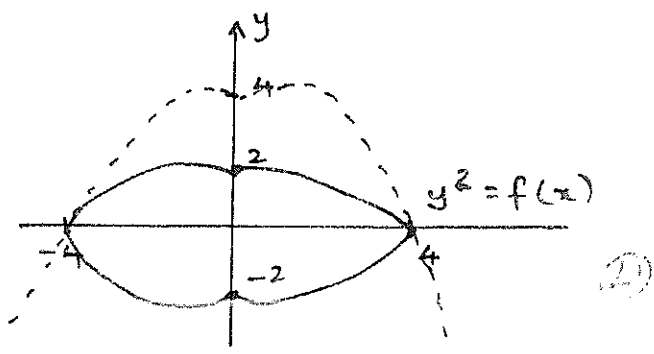
a. i.



ii.



iii.



b. $f(x) = \frac{\ln x}{x}$

i. $y = \frac{\ln x}{x}$

$$y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$$

$$= \frac{1 - \ln x}{x^2} = 0$$

$$1 - \ln x = 0$$

$$\ln x = 1$$

$$x = e$$

$$y(e) = \frac{\ln e}{e} \quad \therefore \text{stat pt } (e, \frac{1}{e})$$

$$= \frac{1}{e}$$

$$y' = \frac{1 - \ln x}{x^2}$$

$$y'' = \frac{x^2 \cdot -\frac{1}{x} - 2x(1 - \ln x)}{x^4}$$

$$= \frac{-x - 2x(1 - \ln x)}{x^4}$$

$$= \frac{-1 - 2(1 - \ln x)}{x^3}$$

$$= \frac{-3 + 2\ln x}{x^3}$$

$$y''(e) = \frac{-3 + 2\ln e}{e^3} < 0 \quad \therefore \text{Max } (e, \frac{1}{e})$$

Now, $y'' = 0 \quad \therefore \frac{-3 + 2\ln x}{x^3} = 0$

$$2\ln x = 3$$

$$\ln x = \frac{3}{2}$$

$$x = e^{3/2}$$

$\therefore$ Poss. pt of inf. at $x = e^{3/2}$

Check neighbourhood

$$y''(e^{3/2} - \epsilon) < 0$$

$$y''(e^{3/2}) = 0$$

$$y''(e^{3/2} + \epsilon) > 0$$

$\therefore$ Sign change

$$\therefore y(e^{3/2}) = \ln e^{3/2}$$

$$= \frac{3}{2} \div e^{3/2}$$

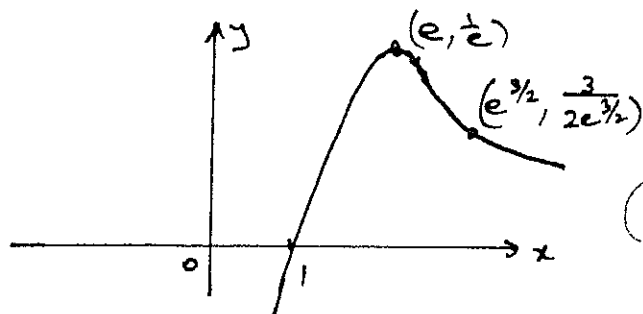
$$\therefore \text{Pt of inf. } (e^{3/2}, \frac{3}{2e^{3/2}})$$

Now, $y = \frac{\ln x}{x}$ DBC at $x=0$
 $\therefore$ Asymp. at $x=0$

Also, $x \rightarrow \infty \therefore \frac{\ln \infty}{\infty} \rightarrow 0$

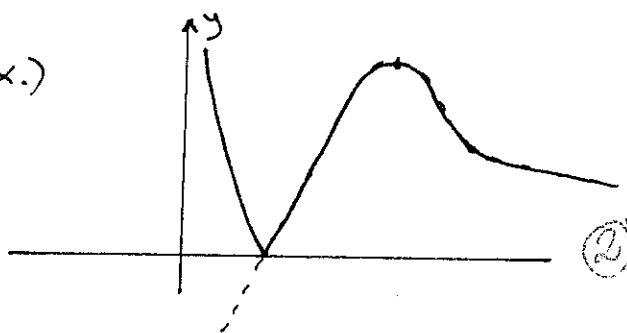
$\therefore$ Asymp at $y=0$

ii.

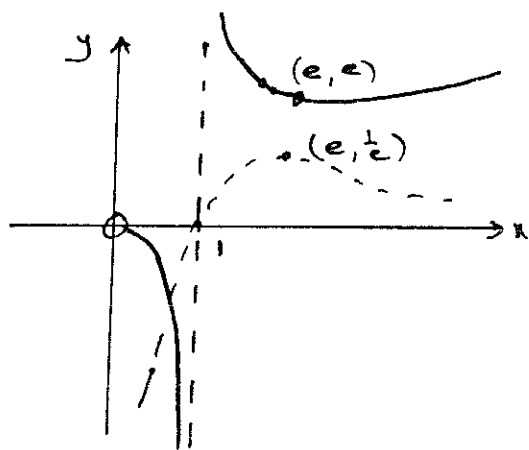


iii.

(α .)



(β)



Question 3

a. i. $z\bar{w} = (5+3i)(-3-2i)$
 $= -15 - 10i - 9i + 6$
 $= -9 - 19i$

ii. $\frac{2}{iw} = \frac{2}{i(-3+2i)}$
 $= \frac{2}{-2-3i} \times \frac{-2+3i}{-2+3i}$
 $= \frac{-4+6i}{4+9}$
 $= \frac{-4}{13} + \frac{6i}{13}$

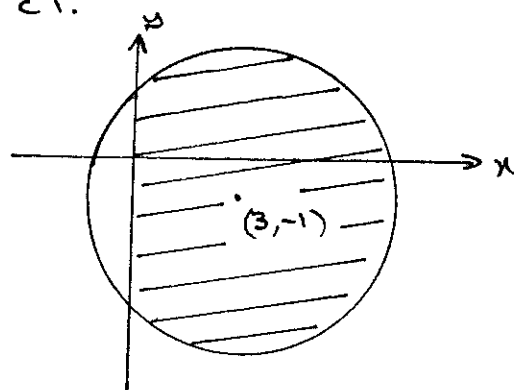
b. i. $\sqrt{3}+i$

Let $z = \sqrt{3}+i$ $|z| = \sqrt{3+1}$
 $= 2$
 $\arg z = \tan^{-1} \frac{1}{\sqrt{3}}$
 $= \frac{\pi}{6}$

$\therefore \sqrt{3}+i = 2 \operatorname{cis} \frac{\pi}{6}$

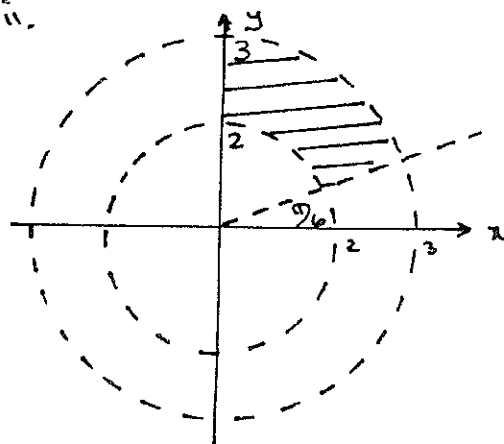
ii. $(\sqrt{3}+i)^{12} = 2^{12} \operatorname{cis} \frac{12\pi}{6}$
 $= 4096 [\cos \frac{2\pi}{1} + i \sin \frac{2\pi}{1}]$
 $= 4096$

c. i.



- $|z-3+i|=5$ is circle with centre $3-i$ and radius 5
- $|z+2|=|z-2|$ is locus of points equidistant from $x=-2$ & $x=2$ i.e. $x=0$ (y axis)

c. ii.



d. $(1+\sqrt{3}i)(1+\sqrt{3}i)$

$= 1 + 2\sqrt{3}i - 3$

$= -2 + 2\sqrt{3}i$

$\therefore (1+\sqrt{3}i)^3 = (1+\sqrt{3}i)(-2+2\sqrt{3}i)$
 $= -2 + 2\sqrt{3}i - 2\sqrt{3}i - 6$
 $= -8$

$$\therefore \omega^3 = \left[\frac{1+\sqrt{3}i}{2} \right]^3$$

$$= \frac{-8}{8}$$

$$= -1$$

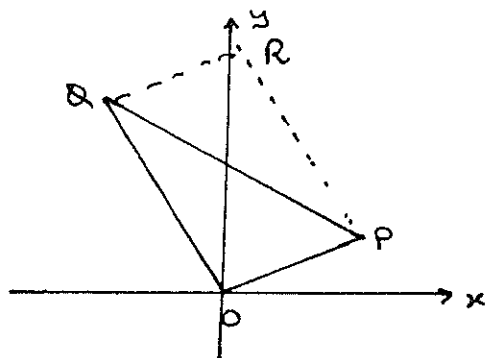
$$\text{ii. } \omega^{16} = (\omega^3)^5 \times \omega$$

$$= (-1)^5 \cdot \omega$$

$$= -\omega$$

$$= \frac{-1-\sqrt{3}i}{2}$$

e.



i.

$$OP = |z_1|, OQ = |z_2|$$

$$PQ = |z_1 - z_2|$$

Now PQ is longest side of $\triangle POQ$

$$\therefore PQ \leq OP + OQ$$

$$\therefore |z_1 - z_2| \leq |z_1| + |z_2|$$

ii. OPRQ is parallelogram

$$\text{iii. } |z_1 - z_2| = |z_1 + z_2|$$

$\therefore$ diagonals of parallelogram equal

$\therefore$ OPRQ is rectangle

$$\therefore \angle POQ = 90^\circ$$

$$\therefore \vec{OQ} = i\vec{OP}$$

$$\therefore z_2 = iz_1$$

$$\therefore \frac{z_2}{z_1} = i$$

Question 4

a. i. Roots are $x = \alpha^2, \beta^2, \gamma^2$

$$\text{If } x = \alpha^2 \therefore \alpha = \sqrt{x}$$

Now as α is root of $x^3 + 5x^2 + 11x + 12 = 0$

$\therefore \sqrt{x}$ is also soln

$$\therefore P(\sqrt{x}) = (\sqrt{x})^3 + 5(\sqrt{x})^2 + 11 = 0$$

$$x\sqrt{x} = -5x - 11$$

Square both sides:

$$x^3 = (-5x - 11)^2$$

$$x^3 = 25x^2 + 110x + 121$$

$$\therefore x^3 - 25x^2 - 110x - 121 = 0$$

ii. $\alpha^2 + \beta^2 + \gamma^2$: sum of roots

$$\therefore 25$$

$$\text{b. i. } P(x) = (x - \alpha)^2 Q(x)$$

$$\therefore P'(x) = 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x)$$

$$= (x - \alpha)[2Q(x) + (x - \alpha)Q'(x)]$$

$$\therefore P'(\alpha) = 0 \therefore x = \alpha \text{ is root of } P'(x)$$

$$\text{ii. } P(x) = x^4 + ax^3 + bx + 21$$

$$P'(x) = 4x^3 + 3ax^2 + b$$

$$P'(1) = 4 + 3a + b = 0$$

$$\therefore 3a + b = -4 \quad \text{--- (1)}$$

$$\text{Also } P(1) = 1 + a + b + 21 = 0$$

$$a + b = -22 \quad \text{--- (2)}$$

$$\text{①} - \text{②}$$

$$2a = 18$$

$$a = 9$$

$$\text{Subs in ② } 9 + b = -22$$

$$b = -31$$

$$\text{iii. } P(x) = x^4 + 9x^3 - 31x + 21$$

$$= (x - 1)^2 \cdot Q(x)$$

$$\text{Now } (x - 1)^2 = x^2 - 2x + 1$$

$$\begin{array}{r} x^2 + 11x + 21 \\ x^2 - 2x + 1 \overline{) x^4 + 9x^3 - 31x + 21} \\ \underline{x^4 - 2x^3 + x^2} \\ 11x^3 - x^2 - 31x \\ \underline{11x^3 - 22x^2 + 11x} \\ 21x^2 - 42x + 21 \\ \underline{21x^2 - 42x + 21} \\ 0 \end{array}$$

$$\therefore (x - 1)^2 (x^2 + 11x + 21)$$

$$= (x - 1)^2 \left(x^2 + 11x + \frac{121}{4} - \frac{121}{4} + 21 \right)$$

$$= (x - 1)^2 \left(\left(x + \frac{11}{2} \right)^2 - \frac{37}{4} \right)$$

$$= (x-1)^2 \left(x + \frac{11}{2} + \frac{\sqrt{37}}{2}\right) \left(x + \frac{11}{2} - \frac{\sqrt{37}}{2}\right)$$

$$= (x-1) \left(x + \frac{11+\sqrt{37}}{2}\right) \left(x + \frac{11-\sqrt{37}}{2}\right) \quad (2)$$

c. As $P(x)$ has real coeff, and $2-i$ is zero $\therefore 2+i$ is also zero

$$\therefore (x-2+i)(x-2-i)$$

$\therefore$ using diff. of 2 squares

$$= [(x-2)+i][(x-2)-i]$$

$$= x^2 - 4x + 4 + 1$$

$$= x^2 - 4x + 5$$

$$\begin{array}{r} x^2 - x - 2 \\ x^2 - 4x + 5 \overline{) x^4 - 5x^3 + 7x^2 + 3x - 10} \\ \underline{x^4 - 4x^3 + 5x^2} \\ -x^3 + 2x^2 + 3x \\ \underline{-x^3 + 4x^2 - 5x} \\ -2x^2 + 8x - 10 \\ \underline{-2x^2 + 8x - 10} \\ 0 \end{array}$$

$$\therefore (x^2 - 4x + 5)(x^2 - x - 2) \quad (3)$$

$$= (x-2+i)(x-2-i)(x-2)(x+1)$$

d. i. $z^5 - 1 = 0$
 $z^5 = 1$

Let $z = \cos \theta + i \sin \theta$

$$\therefore \cos 5\theta + i \sin 5\theta = 1 + 0i$$

$$\therefore \cos 5\theta = 1 \quad \sin 5\theta = 0$$

$$\therefore 5\theta = 0, \pm 2\pi, \pm 4\pi, \pm 6\pi, \dots$$

$$\therefore \theta = \frac{2\pi k}{5} \quad \text{where } k = 0, \pm 1, \pm 2$$

ie $z_1 = \text{cis } 0 = 1$

$$z_2 = \text{cis } \frac{2\pi}{5}$$

$$z_3 = \text{cis } -\frac{2\pi}{5} \quad (2)$$

$$z_4 = \text{cis } \frac{4\pi}{5}$$

$$z_5 = \text{cis } -\frac{4\pi}{5}$$

ii. Now $z_3 = \cos -\frac{2\pi}{5} + i \sin -\frac{2\pi}{5}$

$$= \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}$$

Similarly for z_5

Now $z_1 + z_2 + z_3 + z_4 + z_5 = 0$ using sum of roots $= -\frac{b}{a} = 0$

$$\therefore 1 + \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5} + \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} + \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5} = 0$$

$$\therefore 1 + 2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = 0$$

$$2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = -1$$

$$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2} \quad (2)$$

Question 5

a. i. a. $4x^2 + 9y^2 = 36$

$$\therefore \frac{x^2}{9} + \frac{y^2}{4} = 1$$

ie $a = 3, b = 2$

Using $b^2 = a^2(1 - e^2)$

$$4 = 9(1 - e^2)$$

$$\frac{4}{9} = 1 - e^2$$

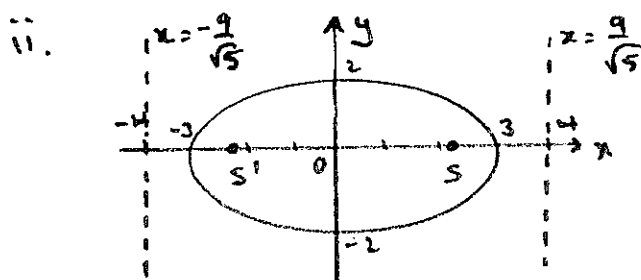
$$e^2 = \frac{5}{9}$$

$$\therefore e = \frac{\sqrt{5}}{3}, e > 0 \quad (1)$$

B. foci: $(\pm ae, 0)$ ie $(\pm \sqrt{5}, 0) \quad (1)$

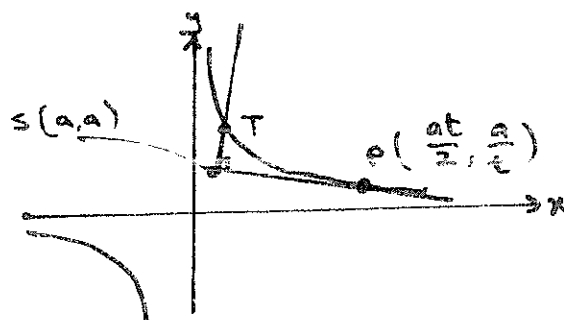
C. Directrices: $x = \pm \frac{a}{e}$
 $= \pm 3 \div \frac{\sqrt{5}}{3}$
 $x = \pm \frac{9}{\sqrt{5}} \quad (1)$

rf. Major axis $= 2a = 2 \times 3 = 6$ units (1)



b. $xy = \frac{1}{2}a^2$

i.



Subs $\left(\frac{at}{2}, \frac{a}{t}\right)$ in $xy = \frac{1}{2}a^2$

$$\text{ie } \frac{At}{2} \cdot \frac{a}{t} = \frac{1}{2}a^2$$

$$\frac{a^2}{2} = \frac{1}{2}a^2 \checkmark$$

$\therefore P$ lies on H

Now, as $xy = \frac{1}{2}a^2$

$$\therefore y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\therefore \text{At } P: y' = -\frac{a}{t} \div \frac{at}{2}$$

$$= -\frac{a}{t} \times \frac{2}{at}$$

$$= -\frac{2}{t^2}$$

$\therefore$ Egn of tangent:

$$y - \frac{a}{t} = -\frac{2}{t^2} \left(x - \frac{at}{2}\right)$$

$$t^2 y - at = -2x + at \quad (2)$$

$$\therefore 2x + t^2 y = 2at \quad (1)$$

ii. grad of normal: $\frac{t^2}{2}$

$$\therefore \text{egn of ST: } y - a = \frac{t^2}{2}(x - a)$$

$$2y - 2a = t^2 x - at^2 \quad (2)$$

$$\therefore t^2 x - 2y = at^2 - 2a \quad (2)$$

iii. If we are to prove a circle,

then we need x^2 and y^2

$$\therefore 2x + t^2 y = 2at \quad (1)$$

$$t^2 x - 2y = at^2 - 2a \quad (2)$$

Square both sides of (1) & (2)

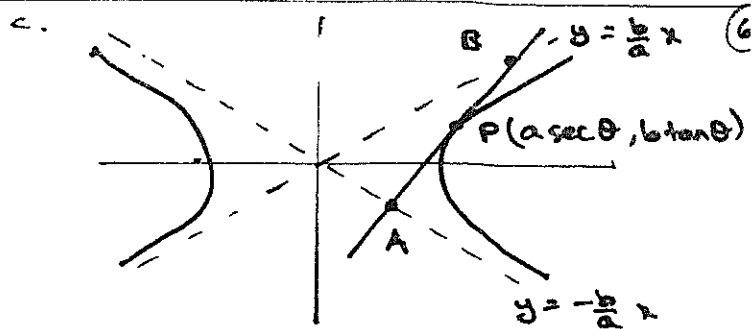
$$\therefore 4x^2 + 4t^2 xy + t^4 y^2 = 4a^2 t^2 \quad (3)$$

$$t^4 x^2 - 4t^2 xy + 4y^2 = a^2 t^4 - 4a^2 t^2 + 4a^2 \quad (4)$$

Now (3) + (4)

$$(t^4 + 4)x^2 + (t^4 + 4)y^2 = a^2(t^4 + 4)$$

$$\therefore x^2 + y^2 = a^2 \therefore \text{circle (2)}$$



$$\therefore \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore \frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{2x}{a^2} \div \frac{2y}{b^2}$$

$$= \frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= \frac{b^2 x}{a^2 y}$$

$$\text{At } P: y' = \frac{b^2}{a^2} \cdot \frac{a \sec \theta}{b \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

$\therefore$ Egn of tangent:

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$bx \sec \theta - ay \tan \theta = ab(\sec^2 \theta - \tan^2 \theta)$$

$$\therefore \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1 \quad (1) \quad (3)$$

ii. Co-ords of B:

Subs $y = \frac{b}{a}x$ in (1)

$$\therefore \frac{x \sec \theta}{a} - \frac{b}{a} \cdot \frac{x \tan \theta}{b} = 1$$

$$x \sec \theta - x \tan \theta = a$$

$$x(\sec \theta - \tan \theta) = a$$

$$x = \frac{a}{\sec \theta - \tan \theta}$$

$$\therefore y = \frac{b}{a} \cdot \frac{a}{\sec \theta - \tan \theta} \quad [\text{ie subs in } y = \frac{b}{a}x]$$

$$= \frac{b}{\sec \theta - \tan \theta}$$

$$\therefore B \left[\frac{a}{\sec \theta - \tan \theta}, \frac{b}{\sec \theta - \tan \theta} \right]$$

Now, coords of A:

Subs $y = -\frac{b}{a}x$ in ①

$$x \sec \theta + x \tan \theta = a$$

$$\therefore x = \frac{a}{\sec \theta + \tan \theta}$$

$$\therefore y = -\frac{b}{a} \cdot \frac{a}{\sec \theta + \tan \theta}$$

$$= \frac{-b}{\sec \theta + \tan \theta}$$

$$\therefore A \left[\frac{a}{\sec \theta + \tan \theta}, \frac{-b}{\sec \theta + \tan \theta} \right]$$

Now, coords of midpoint:

$$x_{mp} = \frac{1}{2} \left[\frac{a}{\sec \theta - \tan \theta} + \frac{a}{\sec \theta + \tan \theta} \right]$$

$$= \frac{1}{2} \left[\frac{a \sec \theta + a \tan \theta + a \sec \theta - a \tan \theta}{\sec^2 \theta - \tan^2 \theta} \right]$$

$$= \frac{1}{2} \left[\frac{2a \sec \theta}{1} \right]$$

$$= a \sec \theta$$

Similarly

$$y_{mp} = \frac{1}{2} \left[\frac{b}{\sec \theta - \tan \theta} + \frac{-b}{\sec \theta + \tan \theta} \right]$$

$$= \frac{1}{2} \left[\frac{b \sec \theta + b \tan \theta - b \sec \theta + b \tan \theta}{\sec^2 \theta - \tan^2 \theta} \right]$$

$$= a \tan \theta$$

$\therefore P(a \sec \theta, a \tan \theta)$ is midpt. ③

Question 6

a i. Let $z = \sqrt[5]{3} \times i$

$$\therefore z^5 = 2 \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)$$

$$= 2 [\cos \pi/6 + i \sin \pi/6]$$

$$= 2 [\text{cis}(\pi/6 + 2\pi k)]$$

$$\therefore z = 2^{1/5} \text{cis} \left(\frac{\pi}{30} + \frac{2\pi k}{5} \right)$$

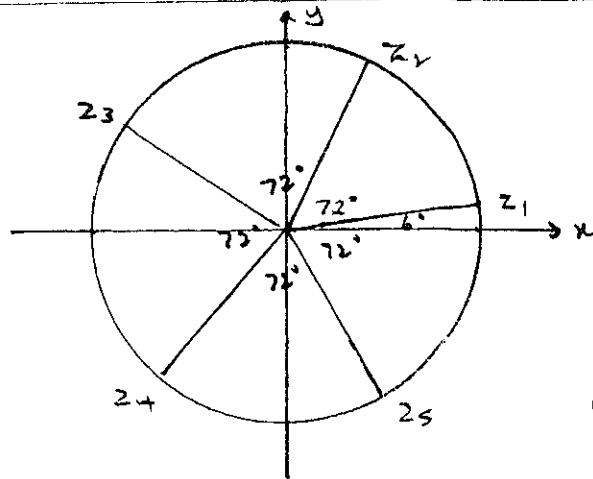
where $k = 0, \pm 1, \pm 2$ ②

ii. Using $\frac{2\pi}{5} = 72^\circ$

$\therefore$ roots differ by 72° .

Now, for $z_1 = 2^{1/5} \text{cis } 6^\circ$

$$(= 2^{1/5} \text{cis } \pi/30)$$



iii. $5 \times \frac{1}{2} ab \sin C$

ie $5 \times \frac{1}{2} \times 2^{1/5} \times 2^{1/5} \sin 72^\circ$

$$= 3.14 u^2$$

b. i. $C_n = \int_0^{\pi/2} \cos^n x \, dx$

$$= \int_0^{\pi/2} \cos x \cdot \cos^{n-1} x \, dx$$

$$u = \cos x \quad v' = \cos x$$

$$u' = (n-1) \cos x \cdot -\sin x$$

$$= -(n-1) \sin x \cos^{n-2} x$$

$$v = \sin x$$

$$\therefore = \sin x \cos^{n-1} x \Big|_0^{\pi/2}$$

$$- \int_0^{\pi/2} -(n-1) \sin x \cos^{n-2} x \sin x \, dx$$

$$= 0 + \int_0^{\pi/2} (n-1) \sin^2 x \cos^{n-2} x \, dx$$

$$= (n-1) \int_0^{\pi/2} (1 - \cos^2 x) \cos^{n-2} x \, dx$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x \, dx - (n-1) \int_0^{\pi/2} \cos^n x \, dx$$

$$= (n-1) C_{n-2} - (n-1) C_n$$

$$\therefore C_n + (n-1) C_n = (n-1) C_{n-2}$$

$$n C_n = (n-1) C_{n-2}$$

$$\therefore C_n = \frac{n-1}{n} C_{n-2}$$

ii. $\int_0^{\pi/2} \cos^4 x \, dx = C_4$

$$\therefore C_4 = \frac{3}{4} C_2 = \frac{3}{4} \left[\frac{1}{2} C_0 \right]$$

$$= \frac{3}{4} \cdot \frac{1}{2} \int_0^{\pi/2} 1 \, dx$$

$$= \frac{3}{8} x \Big|_0^{\pi/2}$$

$$= \frac{3}{8} \left[\frac{\pi}{2} - 0 \right]$$

$$= \frac{3\pi}{16}$$

(1)

c. i. $z = \cos \alpha + i \sin \alpha$

$$z^n = \cos n\alpha + i \sin n\alpha$$

$$\frac{1}{z^n} = z^{-n} = \cos(-n)\alpha + i \sin(-n)\alpha$$

$$= \cos n\alpha - i \sin n\alpha$$

$$\therefore z^n - \frac{1}{z^n} = \cos n\alpha + i \sin n\alpha - (\cos n\alpha - i \sin n\alpha)$$

$$= 2i \sin n\alpha$$

(2)

ii. Using $n=1$

$$\therefore z - \frac{1}{z} = 2i \sin \alpha$$

$$\therefore (z - \frac{1}{z})^3 = 8i^3 \sin^3 \alpha$$

$$= -8i \sin^3 \alpha \quad \text{--- (1)}$$

Also, $(z - \frac{1}{z})^3 = (z - \frac{1}{z})(z - \frac{1}{z})^2$

$$= (z - \frac{1}{z})(z^2 - 2 + \frac{1}{z^2})$$

$$= z^3 - 2z + \frac{1}{z} - z - \frac{2}{z} - \frac{1}{z^3}$$

$$= z^3 - 3z + \frac{3}{z} - \frac{1}{z^3}$$

$$= (z^3 - \frac{1}{z^3}) - 3(z - \frac{1}{z})$$

$$= 2i \sin 3\alpha - 6i \sin \alpha$$

$$= (2 \sin 3\alpha - 6 \sin \alpha) i \quad \text{--- (2)}$$

Now (1) = (2)

$$\therefore -8 \sin^3 \alpha = 2 \sin 3\alpha - 6 \sin \alpha$$

(2)

$$\therefore \sin^3 \alpha = \frac{3}{4} \sin \alpha - \frac{1}{4} \sin 3\alpha$$

iii. Area = $2 \int_0^\pi \sin^3 \alpha \, d\alpha$

$$= 2 \int_0^\pi \left(\frac{3}{4} \sin \alpha - \frac{1}{4} \sin 3\alpha \right) d\alpha$$

$$= \frac{1}{2} \int_0^\pi 3 \sin \alpha - \sin 3\alpha \, d\alpha$$

$$= \frac{1}{2} \left[-3 \cos \alpha + \frac{1}{3} \cos 3\alpha \right]_0^\pi$$

$$= \frac{1}{2} \left[3 - \frac{1}{3} - \left(-3 + \frac{1}{3} \right) \right]$$

$$= \frac{1}{2} \left[2 \frac{2}{3} + 2 \frac{2}{3} \right]$$

$$= 2 \frac{2}{3} \text{ u}^2$$

